

TrendWatch AI — Social Media Intelligence Report

Use Case 2: Brand monitoring on Reddit · Generated 2026-06-09 03:44 UTC

Query: data r/technology Limit: 108 Time: devvit_import

Data: Live Reddit (imported from Devvit app)

Report contents (assignment sections)

1. [Executive briefing \(LLM / fallback\)](#)
2. [Crisis intelligence score](#)
3. [Overview & sentiment](#)
4. [Trend detection](#)
5. [Topic clusters](#)
6. [NER — top entities](#)
7. [Risk keywords & high-impact posts](#)
8. [Post-level analysis](#)
9. [NLP techniques demonstrated](#)
10. [Agent trace & evaluation](#)
11. [Ethics & limitations](#)

1. Executive briefing

Source: rule_based_fallback · RAG + CoT LLM or rule-based fallback

Executive summary

TrendWatch AI analysed 108 Reddit posts. Negative sentiment is 44.4%. Crisis score is 39/100 (Medium).

Main concerns

- Risk keywords appeared 7 times across posts.
- 48 high-engagement posts carry negative sentiment.

Crisis explanation

Risk keywords appeared 7 times across posts. 48 high-engagement posts carry negative sentiment.

Recommended actions

- Monitor high-impact negative threads daily.
- Prepare a factual FAQ for top complaint topics.
- Escalate to customer support if refund/delay keywords spike.

Communication strategy

Address emerging topics: Data / Centers / Center, Data / Breach / With, Data / Centers / Center. Use empathetic, transparent tone; avoid dismissive responses.

Full LLM report (markdown)

Executive Summary

TrendWatch AI analysed 108 Reddit posts. Negative sentiment is 44.4%. Crisis score is 39/100 (Medium).

Main Concerns

- Risk keywords appeared 7 times across posts.
- 48 high-engagement posts carry negative sentiment.

Crisis Explanation

Risk keywords appeared 7 times across posts. 48 high-engagement posts carry negative sentiment.

Recommended Actions

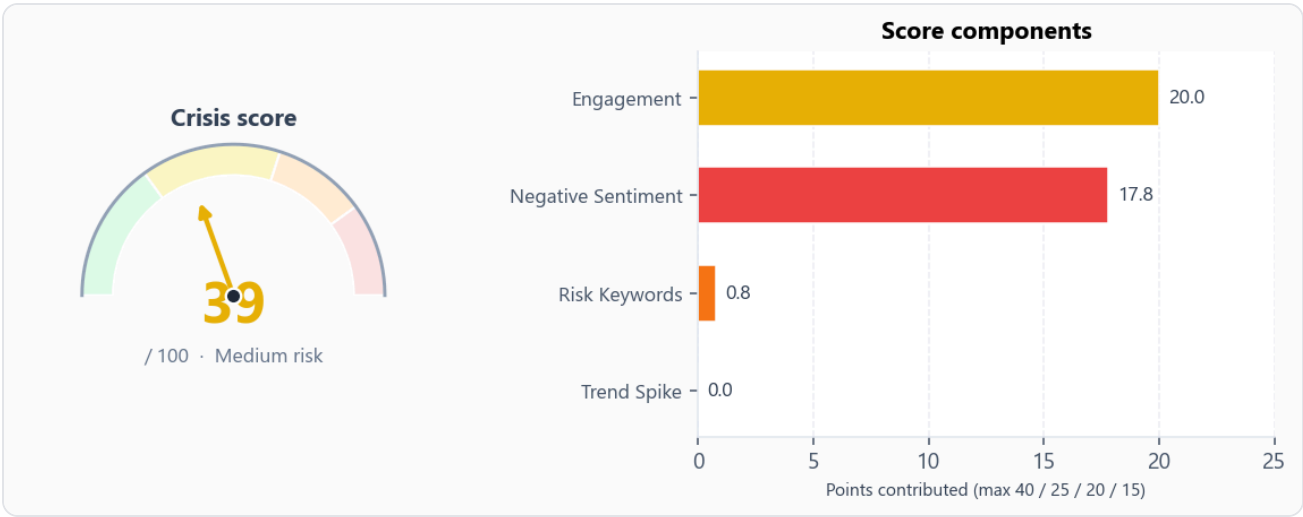
- Monitor high-impact negative threads daily.
- Prepare a factual FAQ for top complaint topics.
- Escalate to customer support if refund/delay keywords spike.

Communication Strategy

Address emerging topics: Data / Centers / Center, Data / Breach / With, Data / Centers / Center. Use empathetic, transparent tone; avoid dismissive responses.

Ethics notice: Analysis uses public Reddit text. Sentiment and crisis scores are algorithmic estimates, not legal or PR advice. Models may reflect language bias; human review is required before brand actions.

2. Crisis intelligence layer



39/100 — Medium risk

Weighted fusion of negative sentiment, risk keywords, high-engagement negative posts, and trend spikes.

Score components

Component	Points
negative_sentiment	17.8
risk_keywords	0.8
engagement	20.0
trend_spike	0.0

Reasons

- Risk keywords appeared 7 times across posts.
- 48 high-engagement posts carry negative sentiment.

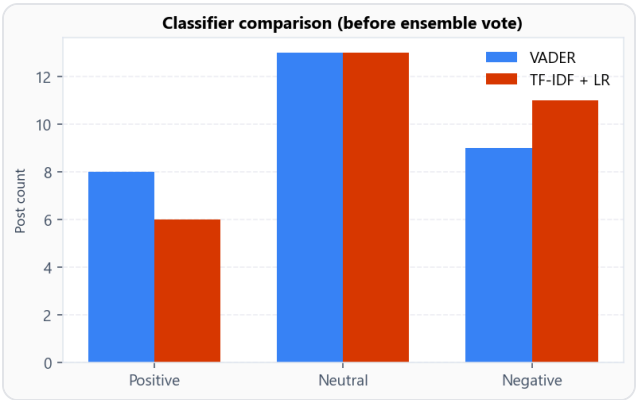
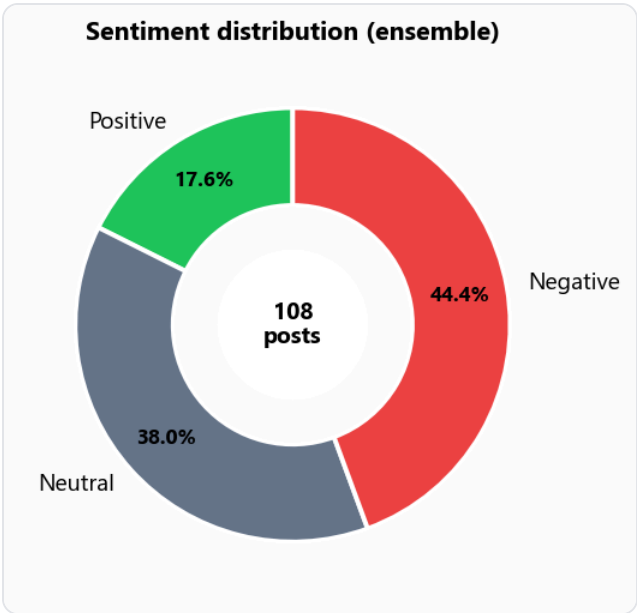
3. Overview & sentiment analysis

108
POSTS ANALYSED

17.6%
POSITIVE

38.0%
NEUTRAL

44.4%
NEGATIVE



Independent classifier outputs before ensemble fusion.

Final labels after VADER + LR ensemble majority vote.

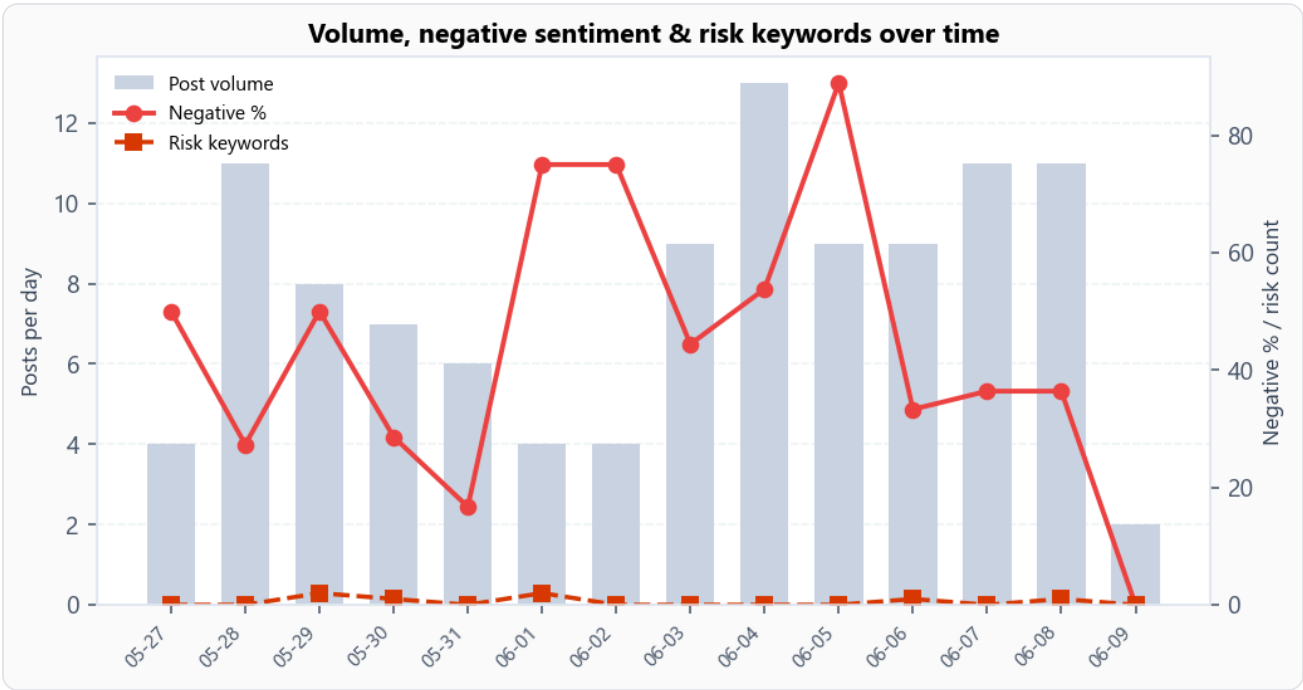
Sentiment distribution (VADER + LR ensemble)

Positive	17.6%
Neutral	38.0%
Negative	44.4%

Label	Count
positive	19
neutral	41
negative	48

Counts: positive=19, neutral=41, negative=48

4. Trend detection



Daily post volume (bars), negative sentiment % (line), and risk keyword counts (dashed).

Spike detected: No — No significant negative sentiment spike detected.

Average negative ratio: 44.0%

Date	Posts	Negative	Neg %	Risk keywords
2026-05-27	4	2	50.0%	0
2026-05-28	11	3	27.3%	0
2026-05-29	8	4	50.0%	2
2026-05-30	7	2	28.6%	1
2026-05-31	6	1	16.7%	0
2026-06-01	4	3	75.0%	2
2026-06-02	4	3	75.0%	0
2026-06-03	9	4	44.4%	0
2026-06-04	13	7	53.8%	0
2026-06-05	9	8	88.9%	0

Date	Posts	Negative	Neg %	Risk keywords
2026-06-06	9	3	33.3%	1
2026-06-07	11	4	36.4%	0
2026-06-08	11	4	36.4%	1
2026-06-09	2	0	0.0%	0

5. Topic clusters (embeddings + KMeans)

Topic 1: Data / Centers / Center

Size: 27 · Avg sentiment: **neutral**

Keywords: data, centers, center, their, power, americans

Representative: Caught on camera: Shelbyville mayor insinuates citizens opposing data centers are poor renters in 'sh***y houses'

Topic 3: Data / Breach / With

Size: 25 · Avg sentiment: **negative**

Keywords: data, breach, with, openai, platform, sensitive

Representative: Signal, DuckDuckGo, and NordVPN threaten to exit Canada if metadata surveillance law passes

Topic 2: Data / Centers / Center

Size: 24 · Avg sentiment: **neutral**

Keywords: data, centers, center, says, meta, build

Representative: AI Data Centers' Water Consumption Breaks 264 Billion Gallons in 2025 as Devastating Drought Hits Nearly 63% of U.S.

Topic 4: Data / Center / Location

Size: 22 · Avg sentiment: **negative**

Keywords: data, center, location, over, centers, massachusetts

Representative: In first, California city overwhelmingly votes to permanently ban datacenters

Topic 0: Data / Center / China

Size: 10 · Avg sentiment: **neutral**

Keywords: data, center, china, boom, power, datacenter

Representative: Kevin O'Leary claims Chinese propaganda is to blame for anti-datacenter backlash
— Administration, industry proponents see foreign plot despite 'scant evidence'

6. Named entities (NER)

Entity	Mentions
kevin (PERSON)	7
utah (GPE)	4
california (GPE)	4
americans (NORP)	4
canada (GPE)	3
china (GPE)	3
millions (CARDINAL)	3
massachusetts (GPE)	2
30 000 (CARDINAL)	2
two (CARDINAL)	2
france (GPE)	2
illinois (GPE)	2
ohio (GPE)	2
new york (GPE)	2
first (ORDINAL)	2

7. Risk keywords & high-impact posts

Risk keyword frequency

Keyword	Count
privacy	3
complaint	1
angry	1
exploit	1
leak	1

Similar post pairs (cosine similarity)

Post A	Post B	Similarity
t3_1tsixo0	t3_1tpxh7z	1.0
t3_1txhq6z	t3_1tvw0po	0.953
Massachusetts votes to pass new privacy rights bill tha	Massachusetts House passes strong data privacy bill ba	0.848
t3_1tqeun2	t3_1tq7ncq	0.799
Kevin O’Leary’s Huge Data Center in Canada Faces a Skep	t3_1twwnk1	0.794

High-impact posts

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
In first, California city overwhelmingly votes to permanently ban datacenters	technology	41938	1360	negative	—	17.86
AI Data Centers’ Water Consumption Breaks 264 Billion Gallons in 2025 as Devasta...	technology	26028	1333	negative	—	17.36
Caught on camera: Shelbyville mayor insinuates citizens opposing data centers ar...	technology	31041	1081	negative	—	17.33

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
Republicans Claim Anti-Data Center Movement Is a Chinese Psy-Op	technology	18432	1758	negative	—	17.29
A Farmer Donated Land to Turn into a Park. The City Is Building a Massive Data C...	technology	27564	919	neutral	—	17.05

8. Post-level analysis (sample)

Showing up to 30 posts with VADER label, LR label, and extracted risk terms.

Title	Author	Ensemble	VADER	LR	Risk terms
What Investment Data Implies about the AI Transition	BathroomMaximum1721	neutral	neutral	neutral	—
Meta launches 'Workforce Academy' to train workers to build data cente...	marketrent	positive	positive	positive	—
Shipowners pursue floating data centers as Samsung Heavy leads push	self-fix2	negative	neutral	negative	—
Massachusetts votes to pass new privacy rights bill that bans sale of ...	Plastic_Ninja_9014	neutral	neutral	neutral	privacy
Working class neighborhoods are resisting data centers at 5 times the ...	inthetownwhere	neutral	positive	negative	—
Hong Kong data centres outpace global average for carbon footprint: UN...	Logical_Welder3467	neutral	neutral	neutral	—
A Farmer Donated Land to Turn into a Park. The City Is Building a Mass...	MarvelsGrantMan136	neutral	neutral	neutral	—
"We pissed off a lot of people": Giant data center plan cut 50% amid p...	AdSpecialist6598	negative	negative	negative	—
Meta So Desperate for Compute That It's	IKeeptLayingAround	negative	negative	negative	—

Title	Author	Ensemble	VADER	LR	Risk terms
Building "Data Centers" That A...					
Majority of US's new AI datacenters to be built on drought-hit land	deraser	neutral	neutral	neutral	—
Amazon engineers in Seattle slam employer for building AI data centers...	Krankenirate	negative	negative	negative	—
OpenAI and Anthropic's next lock-in play: Databases of coding intent	Logical_Welder3467	positive	positive	positive	—
Space Solar agrees to host Lonestar data storage aboard orbital power ...	Logical_Welder3467	positive	positive	positive	—
Kevin O'Leary's Huge Data Center in Canada Faces a Skeptical Public	joe4942	neutral	neutral	neutral	—
Signal, DuckDuckGo, and NordVPN threaten to exit Canada if metadata su...	rkhunter_	negative	negative	negative	—
AI Data Centers' Water Consumption Breaks 264 Billion Gallons in 2025 ...	Wagamaga	negative	negative	negative	—
Data Centers Flock to France's Abundant Nuclear Electricity	Logical_Welder3467	neutral	neutral	neutral	—
The World's First Underwater Data Center Powered by Wind Opens in Chin...	Logical_Welder3467	neutral	neutral	neutral	—

Title	Author	Ensemble	VADER	LR	Risk terms
Oxford University student data pwned yet again - this time via career ...	rkhunter_	neutral	neutral	neutral	—
Massachusetts has put location data startups on notice	rkhunter_	neutral	neutral	neutral	—
South Africa faces calls to block U.S.-listed firm's entry into Africa...	Logical_Welder3467	negative	negative	negative	—
Can ASEAN's Green Goals Survive Its Data Center Boom?	Logical_Welder3467	neutral	neutral	neutral	—
Illinois joins Ohio in ordering pause on data center tax credits	Plastic_Ninja_9014	positive	positive	neutral	—
OpenAI unveils Lockdown Mode to protect sensitive data from prompt inj...	ChestEducational2258	negative	negative	negative	—
Nashville Zoo tries to halt data center project next door over animal ...	Shogouki	positive	positive	positive	—
New York could become first state to temporarily ban large data center...	marketrent	negative	negative	negative	—
Industry coalition urges U.S President administration to take urgent a...	ControlCAD	negative	negative	negative	—
Massachusetts House passes strong data privacy bill banning sale of c...	sr_local	positive	positive	positive	privacy

Title	Author	Ensemble	VADER	LR	Risk terms
Utah residents sue officials over Kevin O’Leary data center plan	esporx	neutral	neutral	neutral	—
Meta Silently Added Face-Recognition Code for Its Smart Glasses to Mil...	Hrmbee	positive	positive	positive	—

9. NLP techniques demonstrated (rubric)

Minimum **3 basic** + **3 advanced** techniques — implementation mapping for your written report.

Basic techniques

- **Preprocessing** (tokenize, stem, lemmatize, URL removal)
- **Named entity recognition** (spaCy / NLTK)
- **Part-of-speech tagging**
- **TF-IDF + Logistic Regression classification**
- **Sentiment ensemble** (VADER + LR majority vote)
- **Embedding-based topic clustering** (KMeans)
- **Regex information extraction** (hashtags, mentions, risk lexicon)

Advanced techniques

- **Foundation LLM** (OpenAI-compatible briefing)
- **Retrieval-augmented generation** (top-k posts)
- **Prompt engineering** (structured analyst template)
- **Chain-of-thought prompting**
- **LLM-as-judge / automated evaluation**
- **Ensemble methods** (sentiment + multi-signal crisis)
- **Agentic ReAct-style orchestration**
- **Ethics & fairness disclaimers**

10. Agent orchestration & evaluation

Agent status: completed

ReAct-style trace

Step	Thought	Action	Observation
1	Need Reddit posts about the brand/topic.	collect_posts	Starting data collection.
2	Collected 108 posts.	collect_posts	Received 108 posts.
3	Raw text must be normalized for NLP.	preprocess	Running tokenizer pipeline.
4	Preprocessing complete.	preprocess	108 posts prepared.
5	Extract hashtags, mentions, risk keywords.	extract	Regex IE.
6	Run NER and POS for entity-aware monitoring.	ner_pos	spaCy/NLTK.
7	Classify sentiment with VADER + LR ensemble.	sentiment	Ensemble vote.
8	Cluster topics with embeddings.	topics	KMeans clustering.
9	Compute temporal trends.	trends	Daily aggregation.
10	Score crisis risk from sentiment + engagement + trends.	crisis	Crisis formula.
11	Retrieve relevant posts for grounded LLM briefing (RAG).	rag	Query=data
12	Generate executive briefing with CoT LLM.	briefing	LLM or fallback.

Automated evaluation

Sentiment agreement (VADER vs LR): 90.7% — VADER and LR agree on 98/108 posts (91%). Use manually labelled gold set for formal accuracy in your report.

Briefing judge score: 70/100 (error_fallback)

LLM judge unavailable; use manual rubric in report.

11. Ethics, limitations & references

--- *Ethics notice: Analysis uses public Reddit text. Sentiment and crisis scores are algorithmic estimates, not legal or PR advice. Models may reflect language bias; human review is required before brand actions.*

- Public Reddit text only; no private user data.
- Crisis scores are decision-support estimates, not legal or PR advice.
- Sentiment models can misclassify sarcasm and domain-specific language.
- For quantitative evaluation on labelled data, extend `backend/evaluation.py`.
- Full NLP technique reference: `docs/NLP_TECHNIQUES.md`

TrendWatch AI · Devvit Web + Python FastAPI · [Raw JSON](#)